KAPASA MAKASA UNIVERSITY

CYS 339: Fraud and Detection Technologies

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Duration: 1 Hour 15 Minutes Total Marks: 100

Instructions:

- Answer ALL questions.
- Section A: Multiple Choice (20 marks)
- Section B: Short Answer (40 marks)
- Section C: Case Study & Practical Application (40 marks)

Section A: Multiple Choice (20 Marks)

1. Which of the following is NOT a type of quantitative data analysis?

a) Descriptive Analysis (b) Diagnostic Analysis (c) Contextual Analysis(d) Predictive Analysis

2. Financial ratio analysis in cyber security is primarily used to:

a) Detect malware signatures b) Measure ROI on security investments

c) Encrypt sensitive data d) Generate phishing emails

3. A favourable variance in cyber security indicates:

a) Actual costs > Expected costs b) Actual costs < Expected costs

c) No change in costs d) Increased breach likelihood

4. Which function combines data from different tables based on a common key?

a) Filter b) Append c) Join d) Pivot

5. Aging analysis is most effective for detecting:

a) Zero-day exploits b) Dormant account reactivation

c) Encryption failures d) Firewall misconfigurations

6. What is the primary limitation of using averages in fraud detection?

a) They are computationally expensive b) They can be skewed by outliers

c) They only work with qualitative data d) They require manual updates

7. Which technique divides data into homogeneous subgroups for analysis?

a) Regression b) Stratification c) VLOOKUP d) Trend analysis

8. A key disadvantage of filtering in cyber security is:

a) Increased data volume b) Risk of over-filtering and missing subtle attacks

c) Inability to handle numerical data d) High cost of implementation

9. Which of the following is a qualitative data analysis method?

a) Sorting b) Behavioral analysis c) Ratio calculation d) Variance analysis

10. What does a high false-positive ratio in IDS alerts suggest?

a) Effective threat detection b) Poorly tuned detection thresholds

c) Encrypted traffic d) Low attack volume

Section B: Short Answer (40 Marks) Answer concisely (2-4 sentences per question).

11. Explain two advantages and two limitations of using financial ratios in cyber security. (4 marks)

12. How does variance analysis contribute to anomaly detection in cyber security? Provide an example.(4 marks)

13. Differentiate between "join" and "append" functions in data analysis with examples. (4 marks)

14. Why is stratification useful in vulnerability management? (4 marks)

15. Describe a scenario where trend analysis could fail to detect a cyber threat. (4 marks)

16. What are the key challenges of using regression models for predicting cyber threats? (4 marks)

17. How does aging analysis help detect insider threats? (4 marks)

18. Compare quantitative and qualitative data analysis in fraud detection. (4 marks)

19. Explain how pivot tables enhance cyber security reporting. (4 marks)

20. Why might percentages be misleading in cyber security metrics? (4 marks)

Section C: Case Study & Practical Application (40 Marks)

Case Study: Phishing Attack Investigation

Background:

A company's SIEM system flagged a 30% increase in phishing emails over three months. Further analysis revealed the following:

1. 60% of emails targeted finance department employees.
2. Mean time to detect (MTTD) increased from 2 hours to 5 hours.
3. Aging analysis showed dormant accounts were accessed after 9+ months.

Tasks:

21. Quantitative Analysis (15 marks):

    a) Calculate the percentage change in MTTD.
    b) A cyber security start up invests ZMW 60,000 per year on security controls. Prior to implementation, annual losses from cyber-attacks averaged ZMW 250,000. After deployment, losses dropped to ZMW 70,000. Calculate the Return On Security Investment (ROI)
    c) Propose a statistical method to predict future phishing spikes.

22. Qualitative Analysis (15 marks):

    a) Analyse behavioural patterns suggesting insider involvement.
    b) Recommend two contextual factors to investigate (e.g., employee training, email templates).

23. Mitigation Strategy (10 marks):

    a) What would you recommend as a mitigation strategy so as to reduce phishing in emails.